2 Variable K-Maps

1. What is a Karnaugh Map?

Theory: There are two main ways to minimize Boolean expressions: using the Laws of Boolean Algebra, or using Karnaugh Maps (K-Maps).

- A K-Map is a visual method to simplify Boolean algebra expressions without needing to manually apply algebraic theorems or rules.
- K-Maps rely entirely on **Gray Code**. In standard binary, stepping from `01` to `10` changes two bits at once. In Gray Code (`00, 01, 11, 10`), adjacent cells differ by only a single bit.
- This 1-bit difference ensures that when we group cells together, variables that change state can be eliminated.

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2. The 2-Variable K-Map

Theory: A 2-variable K-Map is drawn for a Boolean expression consisting of two variables (A and B). The number of cells is calculated as 2^n (where n is the number of variables), so a 2-variable map has $2^2 = 4$ cells.

2-Variable Grid Layout:

$A \backslash B$	0	1
0	m_0	m_1
1	m_2	m_3

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3. SOP Simplification using K-Maps (Grouping 1s)

Theory:

For Sum of Products (SOP), we place a `1` in the cell for every minterm present in the function. We then group adjacent `1`s.

Example 1: Standard SOP

Simplify $f(A, B) = \Sigma m(1, 2, 3)$.

The Map:

$A \backslash B$	$0(\bar{B})$	$1(B)$
$0(\bar{A})$	0	$\mathbf{1}$
$1(A)$	$\mathbf{1}$	$\mathbf{1}$

- **Group 1 (Vertical):** Cells m_1 and m_3. The A variable changes, so it is eliminated. We are left with B.
- **Group 2 (Horizontal):** Cells m_2 and m_3. The B variable changes, so it is eliminated. We are left with A.
- **Simplified Expression:** $F(A, B) = A + B$.

Example 2: Using "Don't Cares"

Simplify $F(A, B) = \Sigma m(0, 2) + d(3)$. *(Note: 'Don't Cares' are marked with an 'X' and can be grouped with 1s if it helps make a larger group!)*

The Map:

$A \backslash B$	$0(\bar{B})$	$1(B)$
$0(\bar{A})$	$\mathbf{1}$	0
$1(A)$	$\mathbf{1}$	$\mathbf{X}$

- **Simplified Expression:** $F(A, B) = \bar{B} + A$.

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4. POS Simplification using K-Maps (Grouping 0s)

Theory: For Product of Sums (POS), we map the maxterms by placing a `0` in the corresponding cells and grouping the `0`s instead of the `1`s.

Example: Standard POS

Simplify $f(A, B) = \Pi M(1, 2, 3)$.

The Map:

$A \backslash B$	$0(B)$	$1(\bar{B})$
$0(A)$	1	$\mathbf{0}$
$1(\bar{A})$	$\mathbf{0}$	$\mathbf{0}$

- **Group 1:** Cells M_1 and M_3 are grouped.
- **Group 2:** Cells M_2 and M_3 are grouped.
- **Simplified Expression:** $f(A, B) = \bar{A} \cdot \bar{B}$.

5. Examples

1. Simplify: $\bar{A}\bar{B} + \bar{A}B + AB$

- **Minterms:** m_0 (`00`), m_1 (`01`), and m_3 (`11`).
- **The K-Map:**

$A \backslash B$	0	1
0	**1**	**1**
1	0	**1**

- **Group 1 (Top Row):** Groups m_0 and m_1. B changes (from 0 to 1), so it drops. A remains `0`. Result = $\bar{A}$.
- **Group 2 (Right Column):** Groups m_1 and m_3. A changes (from 0 to 1), so it drops. B remains `1`. Result = B.
- **Final Minimized Expression:** $\bar{A} + B$

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2. Simplify: $a\bar{b} + ab$

- **Minterms:** m_2 (`10`) and m_3 (`11`).
- **The K-Map:**

$a \backslash b$	0	1
0	0	0
1	**1**	**1**

- **Group 1 (Bottom Row):** We have a single horizontal pair grouping m_2 and m_3. Looking at the columns, b changes from `0` to `1`, so b is eliminated. Looking at the row, a is constant at `1`.
- **Final Minimized Expression:** a